

## Reducibility In Class Examples and Extra Problems

1. Let  $HALT_{TM} = \{\langle M, w \rangle \mid M \text{ is a TM and } M \text{ halts on input } w\}$ . Show that  $HALT_{TM}$  is undecidable.
2. Let  $E_{TM} = \{\langle M \rangle \mid M \text{ is a TM and } L(M) = \emptyset\}$ . Show that  $E_{TM}$  is undecidable.
3. Let  $REGULAR_{TM} = \{\langle M \rangle \mid M \text{ is a TM and } L(M) \text{ is a regular language}\}$ . Show that  $REGULAR_{TM}$  is undecidable.
4. Let  $EQ_{TM} = \{\langle M_1, M_2 \rangle \mid M_1 \text{ and } M_2 \text{ are TMs and } L(M_1) = L(M_2)\}$ . Show that  $EQ_{TM}$  is undecidable.
5. Consider the problem of determining whether a two-tape Turing machine ever writes a nonblank symbol on its second tape when it is run on input  $w$ . Formulate this problem as a language and show that it is undecidable.
6. Consider the problem of determining whether a two-tape Turing machine ever writes a nonblank symbol on its second tape during the course of its computation on any input string. Formulate this problem as a language and show that it is undecidable.